SOFTWARE CHANGE IMPLEMENTATION REPORT

CR033: Last Login Tracking and Display for Student Accounts

Field	Details
Change Request ID	CR033
Change Request Title	Last Login Tracking and Display for Student Accounts
Student Name	J. Thushany
Student ID	JF24ICT09
Implementation Status	Completed
Date	13 October 2026
Project	Student Management System
Branch	JFICT09/CR033
Commit ID	711484b

1. Change Request Summary

1.1 Business Requirement

The Student Management System required a mechanism to record the most recent successful login time for each user account. This information helps administrators understand when an account was last accessed and distinguish between accounts that have logged in and accounts without a recorded login.

The change was required to store login timestamps in the database and retrieve them for display in the administrator dashboard.

1.2 Existing Behaviour

Before implementation:

- The users table stored user account information, including username, password hash, role, and account timestamps.
- The system did not maintain a dedicated last_login field.
- The administrator dashboard did not display the most recent login time for each student.

- Administrators could not directly identify accounts with no recorded login timestamp through the student list.

1.3 Requested Change

The following functionality was required:

- Add a nullable last_login field to the users table.
- Record the current date and time after a successful login.
- Update the timestamp for the corresponding user account.
- Retrieve the login timestamp through the student data model.
- Display the last login date and time in the administrator dashboard.
- Display Never logged in when no login timestamp is available.
- Preserve the existing authentication, student management, and administrator CRUD functionality.

2. Implementation

2.1 Implementation Summary

CR033 was implemented by updating the database schema, backend authentication logic, user model, student model, and administrator dashboard.

A new last_login column was added to the users table. After successful password verification, the backend updates the corresponding user's login timestamp using the current database time.

The student model retrieves the last_login value from the associated user record. The administrator dashboard formats and displays the timestamp in the student list. If the timestamp is missing or invalid, the dashboard displays Never logged in.

The implementation reuses the existing authentication mechanism and database relationships without requiring a new API endpoint.

2.2 Affected Files

File	Change	Purpose
database/schema.sql	Modified	Added the last_login column to the users table definition.

backend/src/controllers/authController.js	Modified	Updated the login process to record the timestamp after successful password verification.
backend/src/models/userModel.js	Modified	Added the function to update the last login timestamp.
backend/src/models/studentModel.js	Modified	Updated student queries to retrieve the associated user's last login timestamp.
frontend/src/pages/DashboardPage.jsx	Modified	Added the Last Login column and formatted timestamp display, including the Never logged in fallback.

Additional files: None.

2.3 Technical Changes

Database Modification

A nullable DATETIME column was added to the users table:

last_login DATETIME NULL DEFAULT NULL

This allows existing accounts to remain valid even if they have no recorded login timestamp.

The column was added to the existing student_management database.

Login Timestamp Update

The authentication controller was updated to call the user model's timestamp update function after successful password verification.

The database update uses:

UPDATE users SET last_login = NOW() WHERE user_id = ?

The user ID identifies the account whose login time must be updated.

Failed password verification does not execute the timestamp update.

Student Data Retrieval

The student model was updated to retrieve the last_login field from the related user record using a join between the students and users tables.

This makes the timestamp available to the administrator dashboard alongside the existing student information.

Dashboard Display

The administrator dashboard now includes a Last Login column in the student table.

The timestamp is formatted into a readable date and time. Missing or invalid values are displayed as Never logged in.

The existing student management operations were retained.

2.4 Implementation Logic

1. A user submits the login form with a username and password.
2. The backend retrieves the corresponding user account.
3. The submitted password is verified against the stored password hash.
4. If authentication succeeds, the backend updates the user's last_login timestamp.
5. The normal authentication process continues.
6. The student model retrieves the timestamp associated with each student.
7. The administrator dashboard displays the Last Login column.
8. If the timestamp is missing or invalid, Never logged in is displayed.
9. Existing student management functionality remains available.

2.5 Error and Data Handling

The implementation includes the following considerations:

- The last_login field accepts NULL for accounts without a recorded login.
- The timestamp update is performed only after successful password verification.
- The update targets the correct account using user_id.
- Missing or invalid timestamps are handled by the dashboard display function.
- Existing login and student data retrieval operations are retained.

3. Testing and Verification

3.1 Testing Approach

The implementation was verified through manual testing in the development environment.

Testing Types:

- Functional testing
- Database verification
- Positive login testing
- Negative login testing
- UI testing
- Regression testing

Environment: Development

Browser: Google Chrome

Test Role: Administrator and Student

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Verify the last_login column in the database.	The column exists and allows NULL values.	Column verified in phpMyAdmin.	Pass
TC-02	Log in using valid administrator credentials.	The administrator's login timestamp is updated.	admin timestamp recorded as 2026-10-13 09:58:54.	Pass
TC-03	Log in using valid student credentials.	The student's login timestamp is updated.	REG001 timestamp recorded as 2026-10-13 09:45:12.	Pass
TC-04	Inspect an account with no recorded login.	The database value remains NULL until a successful login is recorded.	NULL values observed for several accounts.	Pass
TC-05	Verify the Last Login column in the	The dashboard displays the login timestamp.	To be confirmed through dashboard UI testing.	Pending

	administrator dashboard.			
TC-06	Verify the Never logged in display.	Accounts without a valid timestamp display Never logged in.	To be confirmed through dashboard UI testing.	Pending
TC-07	Attempt login with an incorrect password.	Login is rejected and the timestamp is not updated by the failed attempt.	To be confirmed through negative login testing.	Pending
TC-08	Verify existing student management functionality.	Existing student operations continue to work.	To be confirmed through regression testing.	Pending

3.3 Test Summary

Test Result	**Count**
Total Test Cases	8
Passed	4
Pending	4
Failed	0

Overall Test Result Partially Verified

The database column and successful-login timestamp updates were verified. The remaining UI, negative-login, and regression tests must be completed before declaring all tests passed.

4. Acceptance Criteria Verification

ID	**Acceptance Criterion**	**Verification Method**	**Status**
AC-001	The users table contains a nullable last_login field.	Database structure inspection	Met
AC-002	Successful login updates the correct user's timestamp.	SQL query result	Met
AC-003	Failed login does not update the timestamp.	Negative login test	Pending

AC-004	The student model retrieves the login timestamp.	Backend and dashboard verification	Implemented; UI verification pending
AC-005	The administrator dashboard includes a Last Login column.	Dashboard UI test	Pending
AC-006	Accounts without a timestamp display Never logged in.	Dashboard UI test	Pending
AC-007	Existing authentication functionality is preserved.	Login regression testing	Pending
AC-008	Existing student management functionality remains available.	CRUD regression testing	Pending

5. Evidence

A maximum of two relevant screenshots should be attached.

Evidence 01 – Database Last Login Tracking

Related: AC-001, AC-002

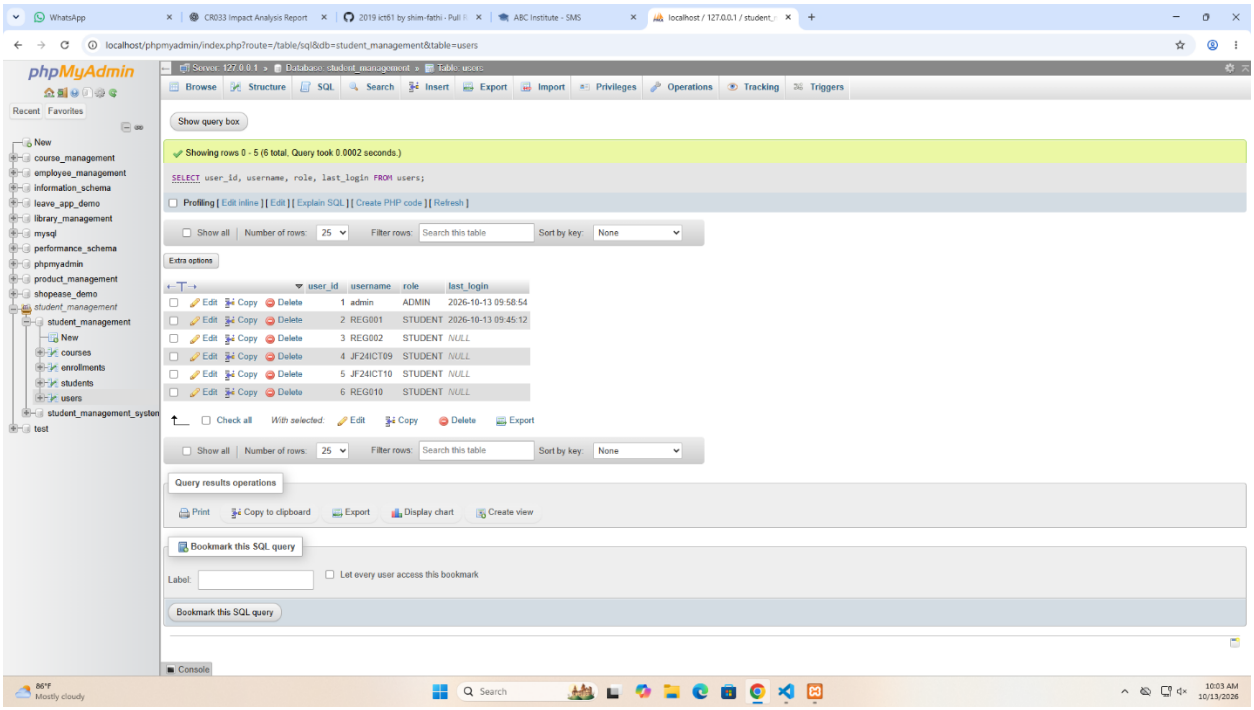


Figure 1: phpMyAdmin showing the query result for user_id, username, role, and last_login in the student_management database.

The screenshot should show the recorded timestamps for admin and REG001, together with NULL values for accounts without a recorded login.

Evidence 02 – Administrator Dashboard Last Login Display

Related: AC-004, AC-005, AC-006

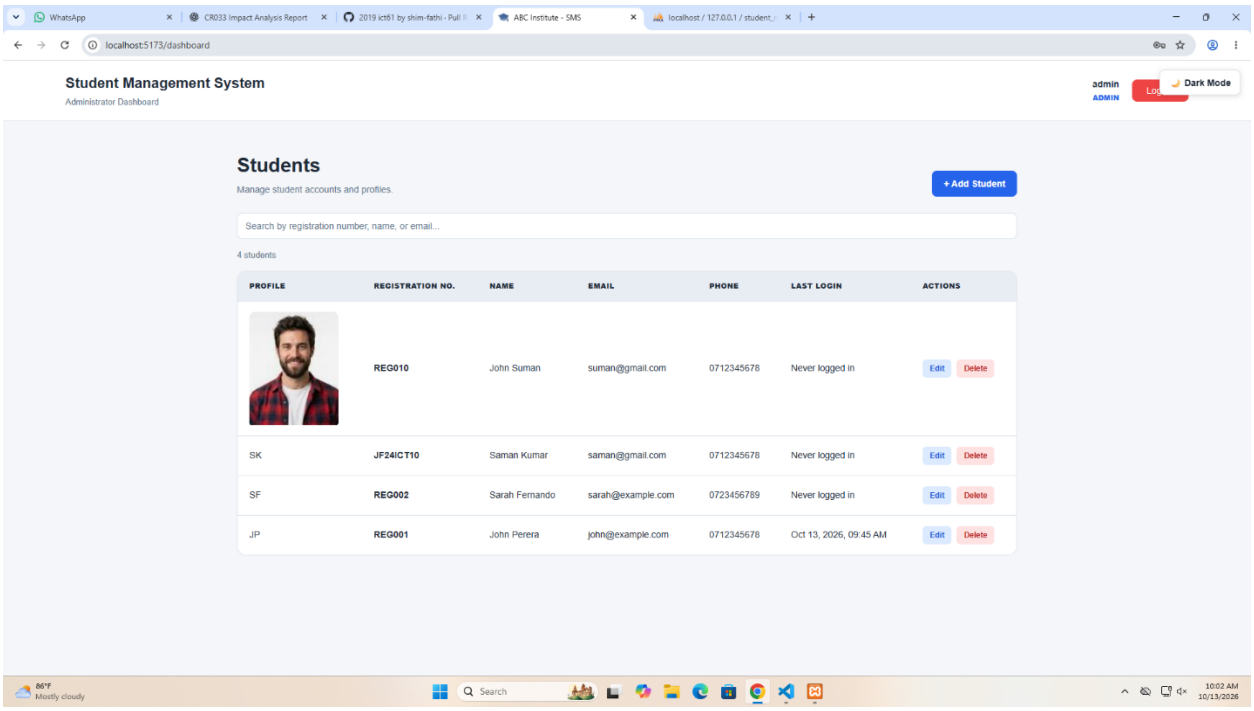


Figure 2: Administrator dashboard showing the student table with the Last Login column, readable timestamps, and the Never logged in fallback where applicable.

Capture this screenshot after completing the dashboard verification.

6. Final Result

Item	Result
Requirements	Partially verified
Implementation	Completed
Database Verification	Passed
Successful Login Timestamp Update	Verified
Dashboard Display	Pending verification
Testing	Partially verified
Acceptance Criteria	Partially verified
Existing CRUD	Pending regression verification
Final Status	Pending final UI and regression testing

6.1 Final Summary

CR033 was implemented to introduce last login tracking and display for student accounts in the Student Management System. The database schema was updated with a nullable last_login field, and the backend authentication process was modified to record the current timestamp after successful password verification.

The user model provides a timestamp update function, while the student model retrieves the last login information from the associated user record. The administrator dashboard was updated to include a Last Login column and a fallback display for accounts without a valid timestamp.

Database verification confirmed that the administrator account and the REG001 student account had recorded login timestamps. Several other accounts had NULL values, indicating that no login timestamp had been recorded for those accounts.

The implementation was committed and pushed to the JFICT09/CR033 branch. Commit 711484b records the implementation changes. The working tree was confirmed clean after the push.

The dashboard display, failed-login behaviour, and existing student management functionality must be verified before all testing and acceptance criteria can be marked as complete.

6.2 Issues and Limitations

- Existing accounts retain NULL values until a successful login is recorded.

- Dashboard rendering and the Never logged in fallback require final UI verification.
- Failed-login and existing CRUD regression tests remain to be completed.
- No new API endpoint was introduced for this change.

Student Name: J. Thushany

Student ID: JF24ICT09

Date: 13 October 2026

Signature: _____

~ END OF REPORT ~